HYPERXITE 11

[Controls]

Graphical User Interface

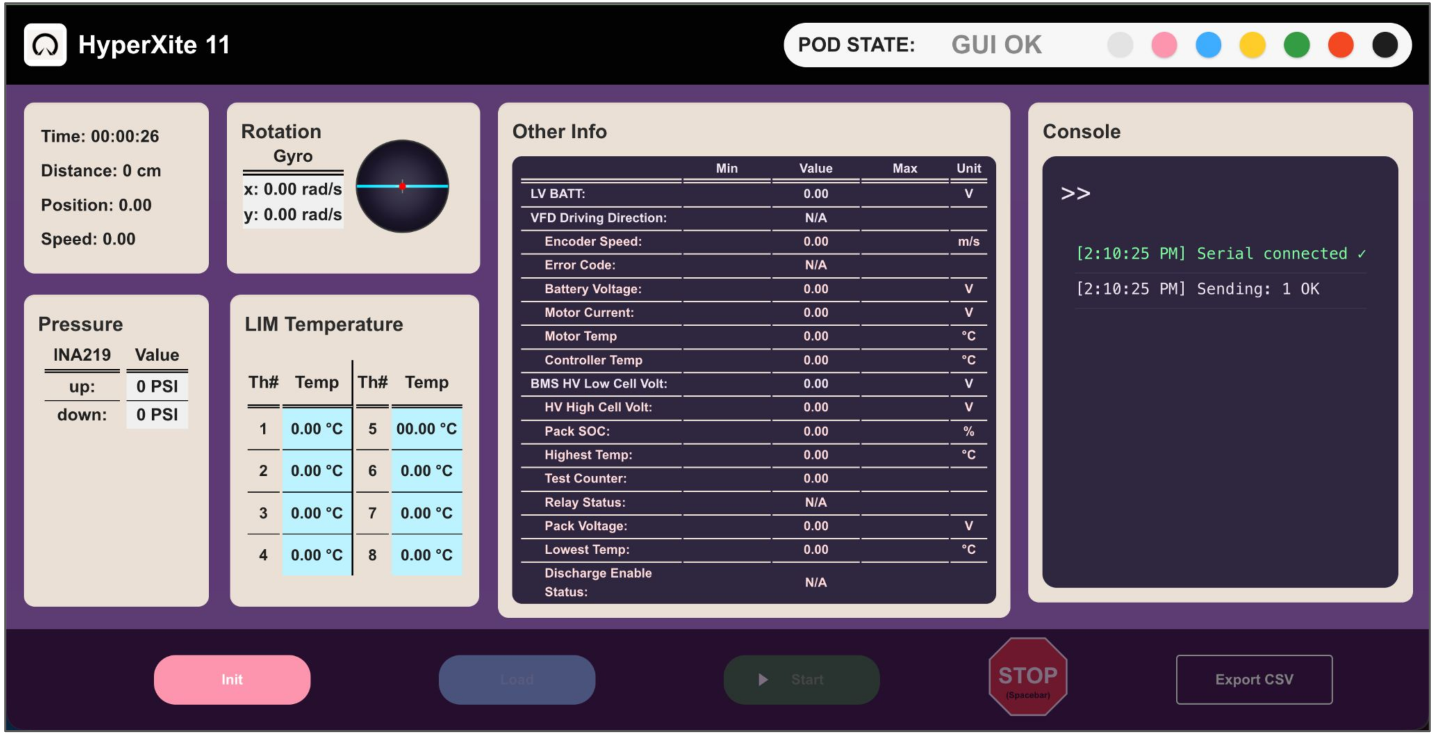

Figure 1. GUI Design

Key Analyses

- Choice of microcontroller: STM32 Nucleo-144 vs. Raspberry Pi 5
  - Pi is great for raw computing power and multitasking (prev. HX design choice)
  - STM32 offers real-time control with low latency
- STM32 offers native CAN and ADC support
  - Less external components/sensors for acquiring data

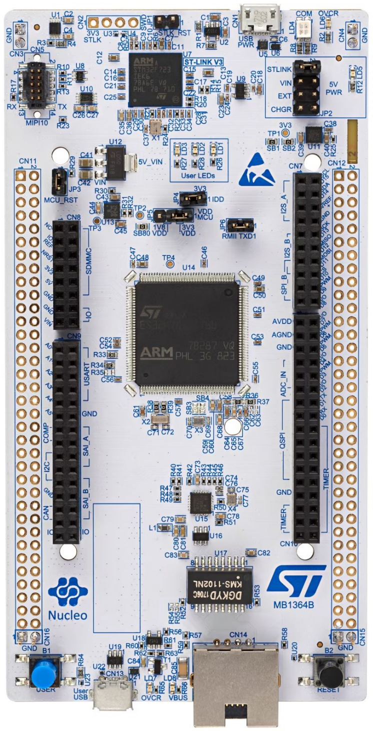

Figure 2. STM32 Nucleo-H753ZI

Constraints and Risks

- System constraints
  - **Limited onboard networking:** STM32 lacks native high-level networking support, constraining direct communication with external systems
  - **Real-time telemetry demands:** The system should support high-speed, low-latency data communication for live GUI visualization
- Risks:
  - **Sensor reliability and detection logic:** Inaccurate sensor readings or missed threshold values by the FSM could prevent timely fault detection, leading to unsafe system behavior

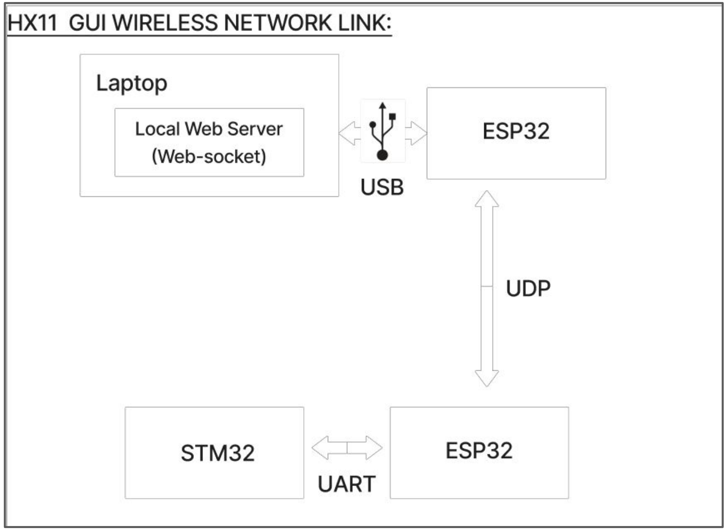

Figure 3. GUI Wireless Network Link

Finite State Machine Design

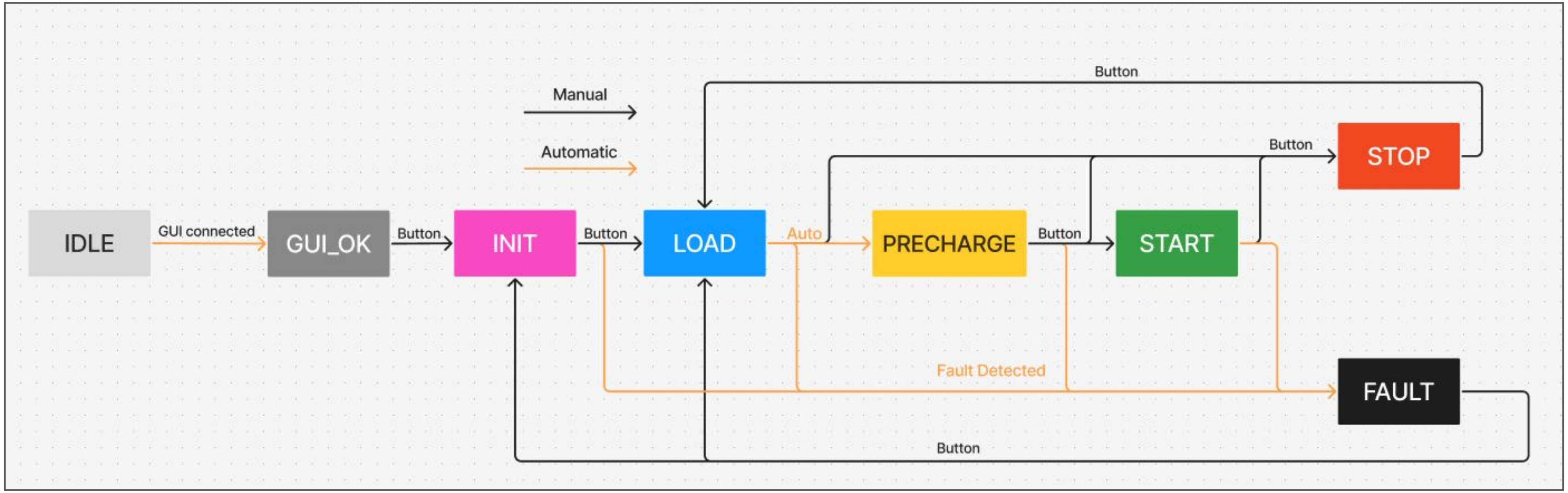

Figure 4: Finite State Machine

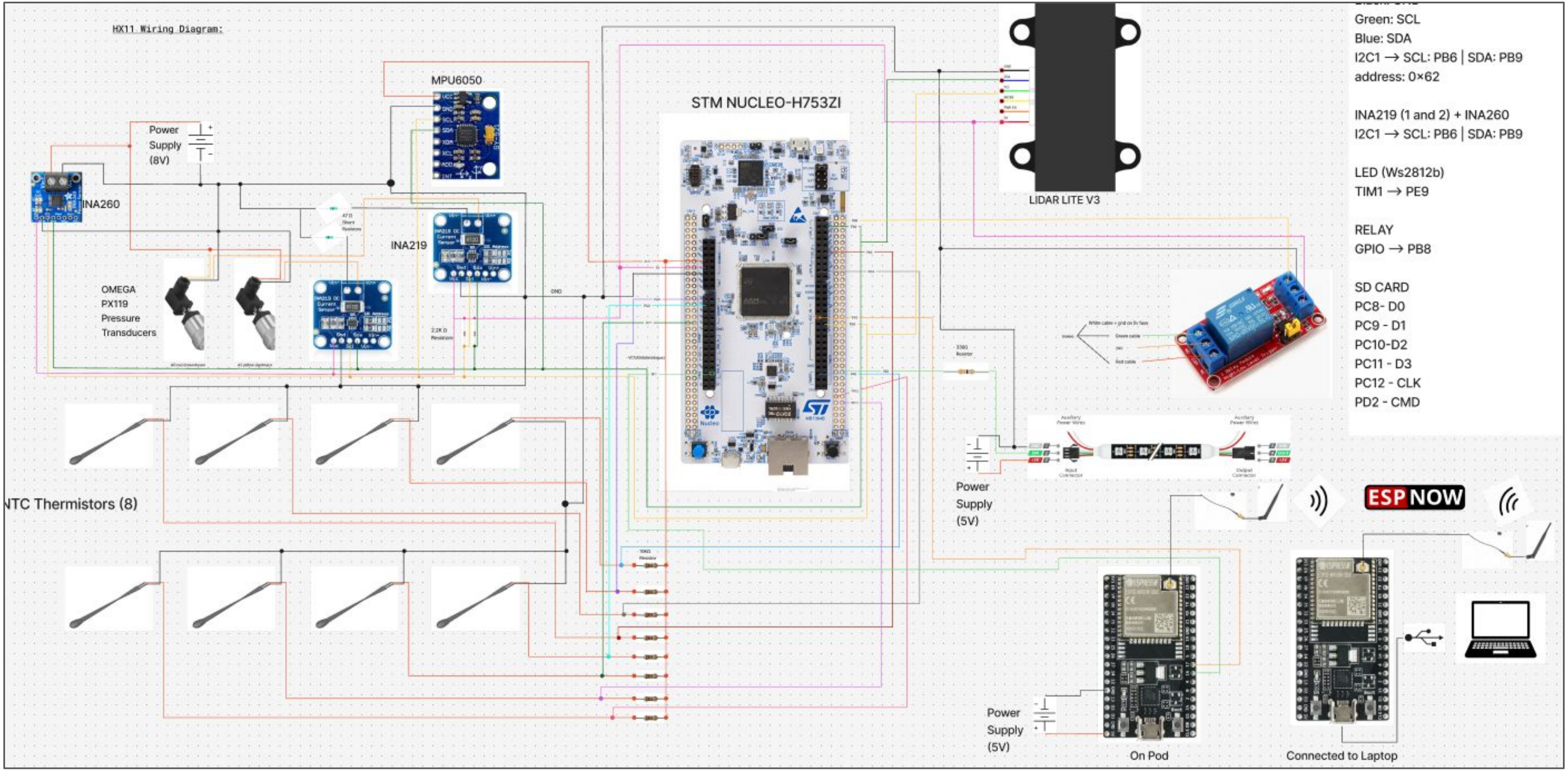

Figure 5. Wiring Diagram

Final Design

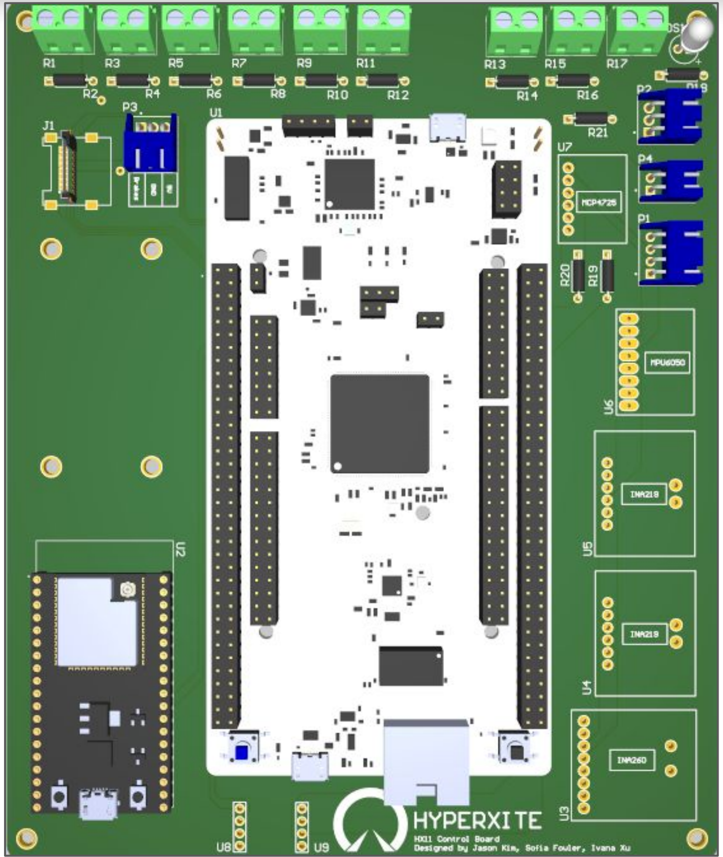

Figure 6: Control Board PCB 3D Model